

Waterfowl Abundance and Precipitation at North Campus Open Space: Temporal and Seasonal Patterns Across Water Years 2020–2025

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2026-06-07

Introduction

Wetlands hold some of the highest concentrations of waterfowl in North America. In our Mediterranean climate, this water is coming almost entirely from winter rain. Precipitation fills wetlands, and often sets habitat and foraging availability for these species. Because of this link, precipitation is oftentimes treated as a basic driver of waterfowl abundance. Literature does not agree with this entirely. Greer et al. (2007) showed that fall flooding can reduce food availability and lower waterfowl survival and reproduction (GREER et al. 2007). Other work points the other way, Kahara et al. (2021) tied drought to reduced habitat quality and smaller populations (Kahara et al. 2021). Together these studies suggest that the effect of precipitation on waterfowl is real, but inconsistent, and likely encompasses other factors like local hydrology, habitat structure, and how a site is managed. Findings from restored coastal wetlands like NCOS are therefore relevant to the broader management of similar systems throughout California and the Pacific Flyway, where precipitation variability and habitat restoration are ongoing concerns.

Our research focuses on waterfowl abundance at North Campus Open Space, a restored coastal wetland at the University of California, Santa Barbara which provides habitat for many bird species, including waterfowl, and its water levels are actively monitored as part of ongoing restoration. Across water years 2020–2025, surveys recorded 19 waterfowl species at the site. We addressed three questions. First, does precipitation affect waterfowl abundance at NCOS? Second, how does waterfowl abundance vary through time at NCOS? Third, are there seasonal patterns shared by waterfowl abundance and precipitation at NCOS? We tested whether the previous month's precipitation predicted total monthly abundance, using a one-month lag to allow time for rainfall to translate into usable habitat. We predicted that higher precipitation in one month would correspond to greater waterfowl abundance in the following month.

Methods

Our study site is North Campus Open Space. This is a restored coastal wetland and estuary that is part of UCSB. The Cheadle Center for Biodiversity and Ecological Restoration (CCEBER) restores and manages this ecosystem. The site provides freshwater wetland, open water and tidal slough habitat used by resident and migratory waterfowl. For our study, we focused on data from water years 2020 to 2025.

Data used for bird abundance comes from CCEBER's `birds.csv` dataset. Waterfowl abundance data came from monthly eBird surveys conducted at NCOS, with one survey per month across the study period. Species were classified by surveyors using the eBird taxonomic grouping and we retained only records in the waterfowl group. We flagged repeat observations to avoid double counting, and filtered for our time frame of 2020-2025. The surveys identified 19 unique waterfowl species.

The data used for precipitation values was obtained from `NOAA_daily_summaries.csv`. This data comes from the National Oceanic and Atmospheric Administration and has weather information for each day. This dataset was refined to only include data within our study period. Then, it was summarized to have the total precipitation value for each month, as the bird surveys are conducted monthly.

We defined the water year as October 1 through September 30 and restricted both datasets to water years 2020-2025. Bird counts were summed by species and by month to produce monthly waterfowl abundance totals. We arranged all of our monthly data chronologically and paired each month's total abundance with the precipitation total from the preceding month to create a one-month lagged precipitation variable.

Statistical analysis. We fit a linear regression with total monthly waterfowl abundance as the response and previous-month precipitation as the prediction. All coding, analysis and figures were produced in Rstudio.

Results

Monthly precipitation at NCOS followed a strongly seasonal pattern across water years 2020–2025, with rainfall concentrated in winter months and dropping to near zero each summer (Figure 1). The highest recorded monthly precipitation occurred in early 2024, while intervening summers were consistently dry. This seasonal structure in precipitation provides the environmental backdrop against which waterfowl abundance patterns are interpreted.

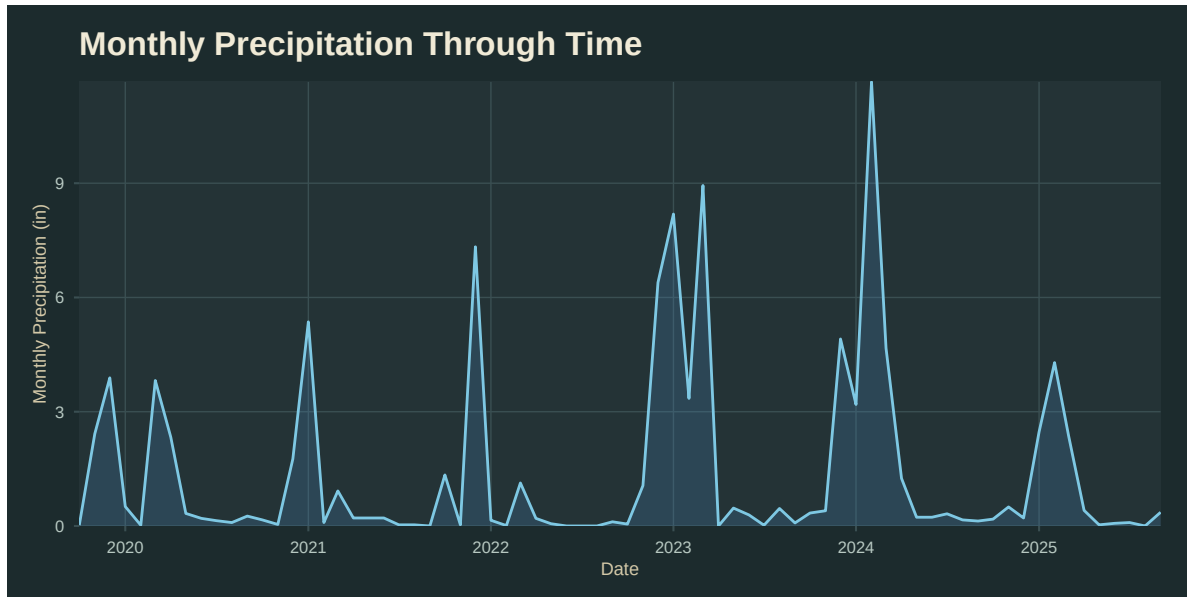


Figure 1: Monthly total precipitation (inches) at NCOS across water years 2020–2025.

Total monthly waterfowl abundance also peaked in winter months and was highest in early 2020, with a general decline in peak abundance observed across the study period (Figure 2). Mallard, Canada Goose, and Green-winged Teal were the dominant species across all years, together comprising the majority of monthly counts. Abundance dropped sharply each summer, with near-zero counts during June through August in most water years. Gaps in the figure reflect months with no survey conducted or no waterfowl observed.

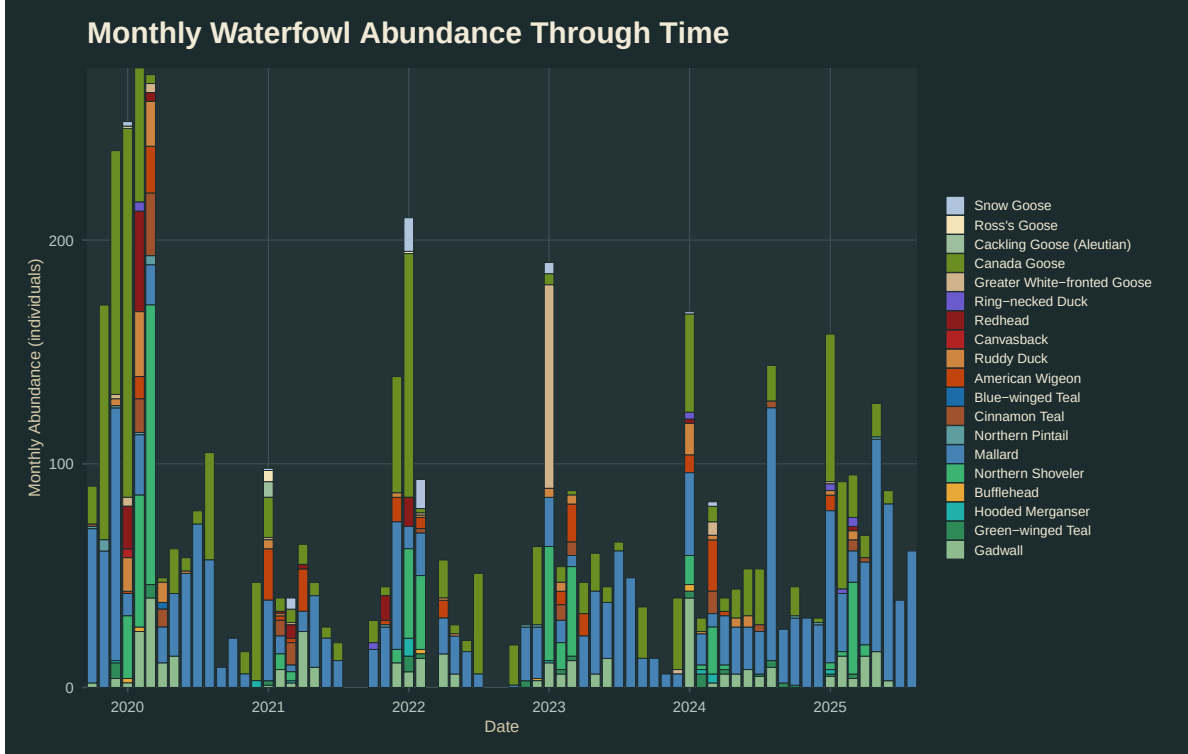


Figure 2: Monthly waterfowl abundance at NCOS across water years 2020–2025, stacked by species.

To address our primary research question, we plotted total monthly waterfowl abundance against the previous month's precipitation and fit a linear regression (Figure 3). Points were heavily clustered near zero precipitation with no clear directional trend, and the regression confirmed no statistically significant relationship between lagged precipitation and waterfowl abundance ($p = 0.085$, $R^2 = 0.046$). While the positive coefficient for lagged precipitation (5.61 individuals per inch) suggests a weak positive trend, it does not meet the conventional threshold for statistical significance ($\alpha = 0.05$), and the model explains less than 4% of the variation in monthly abundance (adjusted $R^2 = 0.031$).

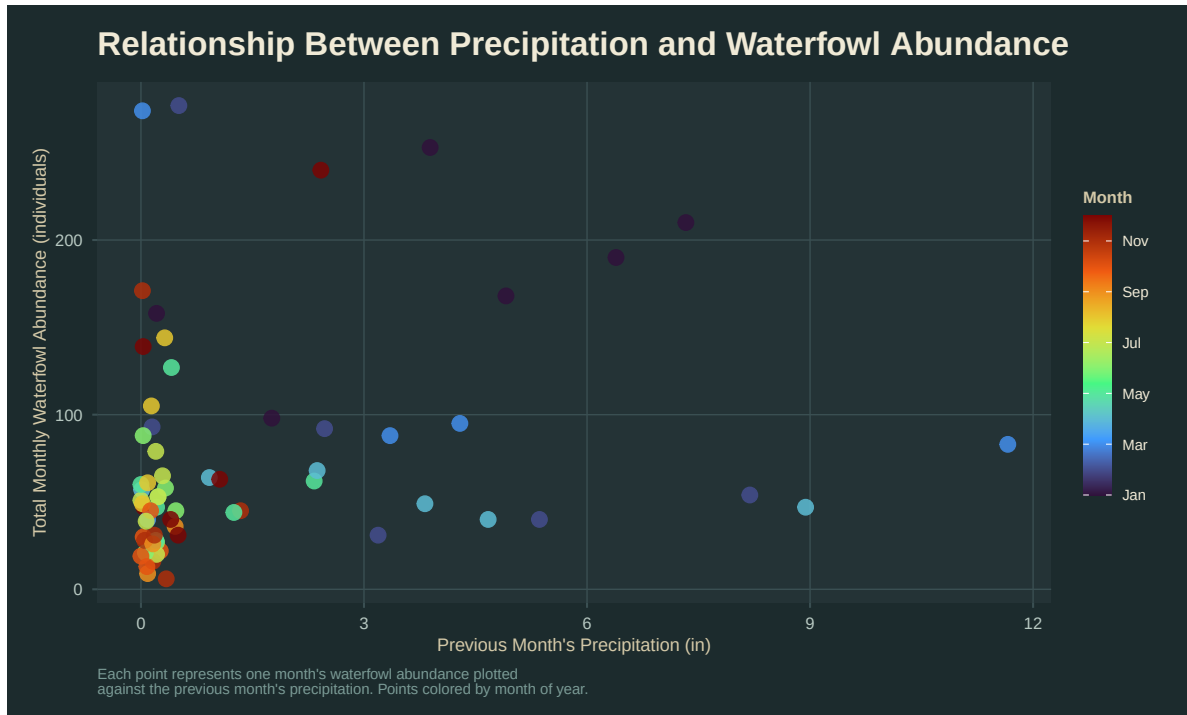


Figure 3: Total monthly waterfowl abundance plotted against the previous month's precipitation at NCOS across water years 2020–2025.

Despite the absence of a direct predictive relationship, Figure 4 reveals that waterfowl abundance and precipitation follow parallel seasonal trajectories when averaged across water years. Both variables peak in winter and decline steadily through spring, summer, and fall. This pattern suggests that while the amount of precipitation in a given month does not predict how many waterfowl will be present the following month, the two variables co-occur seasonally at NCOS, likely because both are driven by the same underlying seasonal forcing rather than one causing the other.

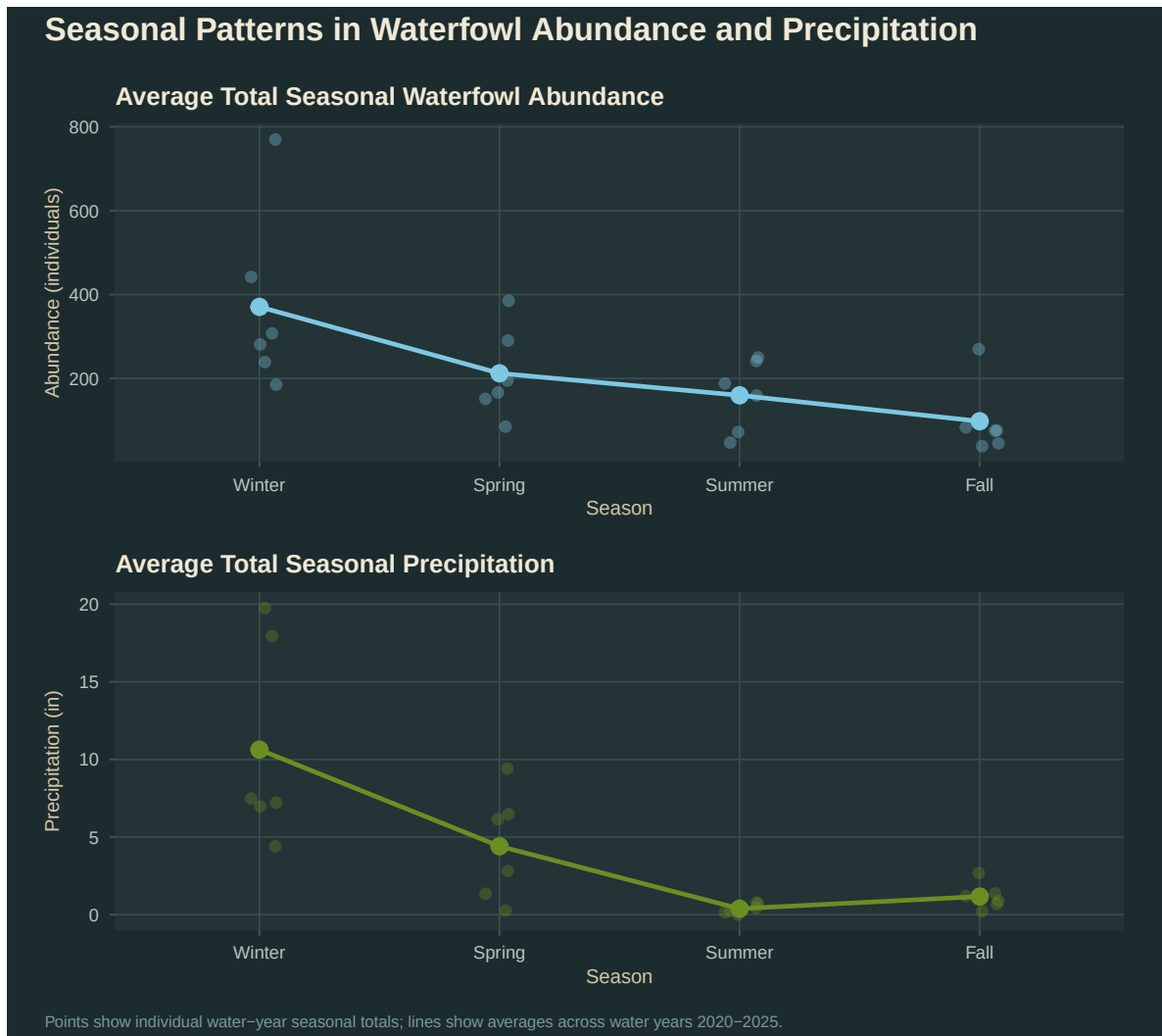


Figure 4: Average total seasonal waterfowl abundance (top) and precipitation (bottom) at NCOS across water years 2020–2025.

Discussion

Our results suggest that precipitation alone did not strongly predict waterfowl abundance at North Campus Open Space (NCOS) from water years 2020-2025. Although it is seen that waterfowl abundance and precipitation follow the same seasonal trends, the direct relationship between previous-month's precipitation totals and waterfowl abundance was not significant. This result indicates that precipitation may not be the main factor controlling waterfowl abundance at NCOS. However, this does not mean that our findings are not meaningful. It

may suggest that local restoration, wetland management, and habitat conditions at NCOS may buffer waterfowl from precipitation-driven fluctuations seen elsewhere.

These results connect to the Bird Community Analysis Report by Karen Stahlheber. This report found that bird patterns at wetland habitats in the UCSB and Goleta area are complex and varying across sites, seasons, and years. The results of their study found that different species of waterfowl preferred different wetland habitats, such as brackish ponds, freshwater marshes, tidal wetlands, and tidal marshes (Stahlheber 2015). This is important for our results because NCOS is a restored wetland system, meaning that habitat structure and the type of wetland habitat available may be just as important as the amount of precipitation. Essentially, waterfowl abundance at NCOS may be more influenced by how precipitation affects wetland conditions than by the precipitation totals alone.

The mixed results that can be found in literature from other ecosystems fits in with our findings. Studies such as Stenzel & Page (2011) found a negative relationship between rainfall and waterbird abundance at a central California estuary Stenzel and Page (2017), while Kahara et al. (2021) found drought reduces waterfowl habitat quality, implying a positive relationship between precipitation and abundance [Kahara et al. (2021)]. Compared to those systems, NCOS may respond differently as it is a restored and partially managed wetland. This could buffer waterfowl from precipitation changes or create habitat conditions that remain usable even when monthly rainfall is low. Rather than contradicting the literature, our null result may validate the inconsistency across studies. The effect of precipitation on waterfowl may depend heavily on local hydrology, habitat structure, and species composition, meaning no universal relationship can be assumed.

Several challenges to data collection may have influenced our results. The first being that bird surveys are difficult to perform. Waterfowl are mobile and may be moving between areas of NCOS. Survey quality may also vary depending on surveyor experience, and climatic conditions. When working with the data, there were some months where surveys did not occur or data was missing. This makes it harder to separate true species absence from limitations in sampling.

Other variables that may influence the patterns seen in this dataset include water depth, wetland area, salinity, vegetation cover, food availability, and temperature. Water variables are collected through gauges, sensors, and YSI EXO1 sondes. Wetland area can be measured through aerial imagery. Vegetation is also measured by the Cheadle Center using transects, quadrat sampling, and individual tree measurements. [citation]. These variables are relevant to our study because precipitation alone doesn't explain the changes to the actual habitat conditions.

Our study encompasses water years 2020-2025. This study period includes seasonal changes, drought years, wet years, and ongoing restoration projects at NCOS. Waterfowl are also migratory birds which can affect their abundance seasonally. Additionally, our study used a one-month lag for precipitation. A longer lag or current-year total precipitation (like in Stenzel & Page 2011) may reveal relationships not captured here. Temporal changes can be seen in our study. Figures 1, 2, and 4 each show seasonal changes in precipitation and waterfowl

abundance. Future research could compare different temporal trends in precipitation to see how the relationship may differ.

Overall, the lack of a significant relationship between the previous month's precipitation and waterfowl abundance suggests that precipitation alone is not a driving factor in waterfowl patterns at North Campus Open Space. Instead, waterfowl abundance is likely influenced by broader seasonal changes and wetland habitat conditions.

References

- GREER, ANDREW K., BRUCE D. DUGGER, DAVE A. GRABER, and MARK J. PETRIE. 2007. "The Effects of Seasonal Flooding on Seed Availability for Spring Migrating Waterfowl." *The Journal of Wildlife Management* 71 (5): 1561–66. <https://doi.org/10.2193/2006-376>.
- Kahara, Sharon N., Daniel Skalos, Buddhika Madurapperuma, and Kaitlyn Hernandez. 2021. "Habitat Quality and Drought Effects on Breeding Mallard and Other Waterfowl Populations in California, USA." *The Journal of Wildlife Management* 86 (1). <https://doi.org/10.1002/jwmg.22133>.
- Stahlheber, Karen A. 2015. *Bird Community Analysis Report 1967-2015*. <https://escholarship.org/uc/item/1sg3v19d>.
- Stenzel, Lynne E., and Gary W. Page. 2017. *Trends in Abundance of Wintering Waterbirds Relative to Rainfall Patterns at a Central California Estuary, 1972–2015*. Western Field Ornithologists. <https://doi.org/10.21199/swb3.13>.